```
begin
    using Symbolics
    using CairoMakie
    using Latexify      # lets Pluto render Symbolics expressions as LaTeX
end
```

Tunnel Lining Under Uniform Ground Pressure — Plane Strain in Polar Coordinates

A long **semicircular tunnel lining** of inner radius R and thickness t (outer radius $R + t$), made of a linear elastic material E, ν, is buried so that the surrounding ground presses on its **outer surface** with a uniform radial pressure p. The inner surface is free. The two feet of the arch rest on **supports** at the level of the centre of the arch. The tunnel is long in the direction of its axis, so every cross-section deforms in the same way and the axial strain is zero: **plane strain** (Sadd §7.1).

The geometry is circular, so this is the first problem of the course that calls for the **polar-coordinate** form of the Airy stress function (Sadd §7.6) and the general **Michell solution** (Sadd §8.3). It is the plane-strain twin of Sadd's Example 8-6, the thick-walled cylinder under uniform pressure, with the extra question of what the supports at the feet of the arch do to that solution.

References: Sadd, *Elasticity*, §7.1 (plane strain), §7.2 and Table 7-1 (plane stress ↔ plane strain), §7.6 (polar coordinate formulation), §8.3.1–8.3.2 (Michell solution, axisymmetric case), Example 8-6 and §8.4.1–8.4.2 (thick-walled cylinder and its limiting cases).

Coordinates. Origin at the centre of the arch, on the line of the supports; x horizontal, y upward; polar angle θ measured from the $+x$ axis, so the right foot is at $\theta = 0$, the crown at $\theta = \pi/2$ and the left foot at $\theta = \pi$. The lining occupies $R \le r \le R + t$, $0 \le \theta \le \pi$. The radial displacement u_r is positive outward and u_θ positive counter-clockwise — so at the right foot u_θ is the *upward* displacement and at the left foot the *downward* one.

Steps:

1. Write the governing equations in polar coordinates: stresses from $\phi(r, \theta)$, the biharmonic equation, plane strain Hooke's law, strain–displacement relations.
2. State the boundary conditions, use symmetry, and be explicit about what the supports are taken to do.
3. Pick the terms of the Michell solution that fit the problem and integrate the general displacement field.
4. Enforce the support conditions, then the pressure conditions, symbolically with `Symbolics.jl`.
5. Recover the stresses (including the out-of-plane σ_z), the strains and the displacements, and compare with the thin-ring formula of strength of materials.
6. Work a numerical example for a concrete lining with `Makie.jl` plots.

1. Governing equations in polar coordinates

With no body forces, an Airy stress function $\phi(r,\theta)$ generates stresses that satisfy the polar equilibrium equations (Sadd 7.6.3) identically (Sadd 7.6.7):

$$\sigma_r = \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2}, \qquad \sigma_\theta = \frac{\partial^2 \phi}{\partial r^2}, \qquad \tau_{r\theta} = -\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial \phi}{\partial \theta}\right)$$

Compatibility again reduces to the biharmonic equation, now built from the polar Laplacian (Sadd 7.6.8):

$$\nabla^4 \phi = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\right)\left(\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2}\right) = 0$$

This equation is the same for plane stress and plane strain (Sadd §7.5): the **stresses** of a traction boundary-value problem do not care which one we are in, and do not depend on E or ν. The difference appears when we go from stresses to strains. **Plane strain** ($\varepsilon_z = \gamma_{rz} = \gamma_{\theta z} = 0$) Hooke's law, inverted (Sadd 7.1.3 and 7.6.2):

$$\varepsilon_r = \frac{1+\nu}{E}[(1-\nu)\sigma_r - \nu\sigma_\theta], \qquad \varepsilon_\theta = \frac{1+\nu}{E}[(1-\nu)\sigma_\theta - \nu\sigma_r], \qquad \gamma_{r\theta} = \frac{2(1+\nu)}{E}\tau_{r\theta}$$

and the out-of-plane normal stress needed to hold $\varepsilon_z = 0$ is

$$\sigma_z = \nu\,(\sigma_r + \sigma_\theta)$$

Equivalently, take the plane stress formulas of Lectures 11–12 and replace $E \to E/(1-\nu^2)$, $\nu \to \nu/(1-\nu)$ (Sadd Table 7-1). Finally, the strain–displacement relations in polar coordinates (Sadd 7.6.1):

$$\varepsilon_r = \frac{\partial u_r}{\partial r}, \qquad \varepsilon_\theta = \frac{1}{r}\left(u_r + \frac{\partial u_\theta}{\partial \theta}\right), \qquad \gamma_{r\theta} = \frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}$$

Two terms are new compared with Cartesian coordinates. The u_r/r in the hoop strain says that a ring which simply moves outward gets longer; the $-u_\theta/r$ in the shear strain is what makes a rigid rotation $u_\theta = \omega_0 r$ strain-free.

2. Boundary conditions — symmetry, and what the supports are taken to do

Outer surface $r = R + t$, for all $0 \le \theta \le \pi$: the ground pushes radially inward with intensity p and (we assume) exerts no shear:

$$\sigma_r(R+t,\theta) = -p, \qquad \tau_{r\theta}(R+t,\theta) = 0$$

Inner surface $r = R$: traction-free,

$$\sigma_r(R,\theta) = 0, \qquad \tau_{r\theta}(R,\theta) = 0$$

The feet $\theta = 0$ and $\theta = \pi$, for all $R \le r \le R + t$. The sketch shows the feet bearing on supports, and we have to decide what those supports do. The choice that makes the problem solvable in closed form is a **smooth, rigid support**: it stops the foot from moving vertically, lets it slide horizontally, and exerts no friction:

$$u_\theta(r,0) = 0, \quad \tau_{r\theta}(r,0) = 0, \qquad\qquad u_\theta(r,\pi) = 0, \quad \tau_{r\theta}(r,\pi) = 0$$

The vertical reaction is **not** prescribed: it is whatever $\sigma_\theta(r,0)$ the solution produces, and we will integrate it afterwards to check that the supports carry the load.

Symmetry — two of them. The problem is mirror-symmetric about the vertical axis $\theta = \pi/2$, so the arch cannot drift sideways as a whole: $u_\theta(r,\pi/2) = 0$. More useful is a second, hidden symmetry. Imagine the **complete ring** $0 \le \theta < 2\pi$ under the same uniform external pressure p. It is symmetric about the horizontal diameter, and on a plane of symmetry the displacement normal to the plane (here the vertical one, u_θ) and the shear traction $\tau_{r\theta}$ both vanish — *exactly* our support conditions. So the upper half of the full ring **is** the arch on smooth supports, with the lower half playing the part of the supports. And a full ring under uniform pressure is **axisymmetric**: nothing can depend on θ. That is what lets us keep only the θ-independent terms of the Michell solution. We will not *assume* axisymmetric displacements, though — the support conditions will enforce it, and in doing so they say something about a term of the general solution that the full ring would never let us see.

What is relaxed, and what it costs

- **Smooth supports.** A real invert, footing or rock bed resists the small inward slide of the feet by friction, or is cast integral with the lining. The difference between the real and the idealised support is a self-equilibrated horizontal traction on the foot; Saint-Venant's principle (Sadd §5.6) confines its effect to within about one lining thickness t of the feet. The hoop stress in the haunches and crown is unaffected.
- **Uniform radial pressure.** The ground pressure on a real tunnel is larger vertically than horizontally (the lateral earth-pressure coefficient is usually below one) and is not purely radial. The non-axisymmetric part needs the $\cos 2\theta$ terms of the Michell solution — the same terms that solve the stress concentration at a hole (Sadd §8.4.3). Here we solve for the axisymmetric part only.
- **Plane strain.** Setting $\varepsilon_z = 0$ requires a longitudinal stress $\sigma_z = \nu(\sigma_r + \sigma_\theta)$ along the tunnel. Far from the portals the surrounding ground supplies it; near a free end it is missing and a Saint-Venant end effect appears (Sadd §7.1 describes the corrective solution).

3. Symbolic setup

Independent variables, problem parameters and the differential operators in r and θ.

```
Differential(θ, 1)
```

```
begin
    @variables r θ
    @variables p R t E ν          # ground pressure, inner radius, lining thickness, Young's modulus, Poisson's ratio
    Dr = Differential(r)
    Dθ = Differential(θ)
end
```

⚠ Attempting to print a symbolic expression in a Pluto notebook. Please run `import Latexify` to enable pretty-printing of symbolic expressions. This warning will only display once.

laplacian_polar

```
laplacian_polar(f)
```

The two-dimensional Laplacian in polar coordinates, $\nabla^2 = \partial^2/\partial r^2 + (1/r)\,\partial/\partial r + (1/r^2)\,\partial^2/\partial\theta^2$ (Sadd 7.6.5).

```
"""
    laplacian_polar(f)

The two-dimensional Laplacian in polar coordinates, ∇² = ∂²/∂r² + (1/r) ∂/∂r + (1/r²) ∂²/∂θ² (Sadd 7.6.5).
"""
laplacian_polar(f) = expand_derivatives(Dr(Dr(f)) + Dr(f)/r + Dθ(Dθ(f))/r^2)
```

biharmonic_polar

```
biharmonic_polar(ϕ)
```

$\nabla^4\varphi = \nabla^2(\nabla^2\varphi)$ in polar coordinates (Sadd 7.6.8).

```
"""
    biharmonic_polar(ϕ)

∇⁴ϕ = ∇²(∇²ϕ) in polar coordinates (Sadd 7.6.8).
"""
biharmonic_polar(ϕ) = simplify(laplacian_polar(laplacian_polar(ϕ)))
```

stresses_polar

```
stresses_polar(ϕ) -> (σr, σθ, τrθ)
```

Stresses from the Airy stress function in polar coordinates (Sadd 7.6.7).

```
"""
    stresses_polar(ϕ) -> (σr, σθ, τrθ)

Stresses from the Airy stress function in polar coordinates (Sadd 7.6.7).
"""
stresses_polar(ϕ) = (simplify(expand_derivatives(Dr(ϕ)/r + Dθ(Dθ(ϕ))/r^2)),
                     simplify(expand_derivatives(Dr(Dr(ϕ)))),
                     simplify(expand_derivatives(-Dr(Dθ(ϕ)/r))))
```

Helpers. `coeffs_in` is the same as in Lecture 12: a condition that must hold *for all* r is equivalent to every coefficient of its polynomial in r vanishing. The integration helper changes: the stresses in polar coordinates are not polynomials in r but **Laurent polynomials** (powers $r^{-2}, r^0, r^2, \ldots$), possibly multiplied by $\ln r$. `∫r` clears the negative powers by multiplying through by r^k, reads off the coefficients, and integrates term by term using $\int r^m dr = r^{m+1}/(m+1)$, $\int r^{-1} dr = \ln r$ and $\int r^m \ln r \, dr = \frac{r^{m+1}}{m+1}\ln r - \frac{r^{m+1}}{(m+1)^2}$.

```
∫r (generic function with 2 methods)
```

```
begin
    coeffs_in(expr, vars...) =
    collect(values(first(Symbolics.polynomial_coeffs(expand(expr), collect(vars)))))

    function ∫r(expr; k = 4)
        ℓ = only(@variables ℓ)                                                  # stands
        in for log(r) while we read coefficients
        e = expand(Symbolics.simplify_fractions(expand(expr) * r^k))
        e = expand(substitute(e, Dict(log(r) => ℓ)))
        d, rem = Symbolics.polynomial_coeffs(e, [r, ℓ])
        isequal(Symbolics.value(rem), 0) || error("expression is not a Laurent
        polynomial in r (times log r)")
        s = 0
        for (mono, co) in d
            m = Symbolics.degree(mono, r) - k
            n = Symbolics.degree(mono, ℓ)
            n ≤ 1 || error("powers of log r above the first are not handled")
            if n == 0
                s += m == -1 ? co*log(r) : co*r^(m + 1)/(m + 1)
            else
                s += m == -1 ? co*log(r)^2/2 : co*(r^(m + 1)/(m + 1)*log(r) - r^(m +
                1)/(m + 1)^2)
            end
        end
        return expand(s)
    end

    ∫r(expr, a, b; k = 4) = expand(substitute(∫r(expr; k), Dict(r => b)) -
    substitute(∫r(expr; k), Dict(r => a)))
end
```

solve_linear_bcs

```
solve_linear_bcs(eqs, unknowns) -> (solution::Dict, n_redundant)
```

Exact Gauss–Jordan elimination on the augmented matrix [A | b] of a linear system that may have more equations than unknowns (same as Lecture 12). Numeric pivots are preferred, so we never divide by a symbolic parameter unless we must. Errors if the system is inconsistent or leaves an unknown undetermined.

```
"""
    solve_linear_bcs(eqs, unknowns) -> (solution::Dict, n_redundant)

Exact Gauss–Jordan elimination on the augmented matrix [A | b] of a linear system that may have
more equations than unknowns (same as Lecture 12). Numeric pivots are preferred, so we never divide by a
symbolic parameter unless we must. Errors if the system is inconsistent or leaves an unknown undetermined.
"""
function solve_linear_bcs(eqs, unknowns)
    isz(a)  = isequal(Symbolics.value(simplify(a)), 0)
    isnum(a) = Symbolics.value(a) isa Number
    lhs = [eq.lhs - eq.rhs for eq in eqs]            # every entry must equal zero
    A   = Symbolics.jacobian(lhs, unknowns)           # linear in the unknowns ⇒ A is free of them
    b   = -substitute.(lhs, Ref(Dict(unknowns .=> 0)))
    M   = Num.(hcat(A, b))
    m, n = size(A)
    row, pivots = 1, Int[]
    for col in 1:n
        cand = [i for i in row:m if !isz(M[i, col])]
        isempty(cand) && continue
        k = findfirst(i -> isnum(M[i, col]), cand)
        i₀ = k === nothing ? cand[1] : cand[k]
        M[row, :], M[i₀, :] = M[i₀, :], M[row, :]
        M[row, :] = simplify.(M[row, :] ./ M[row, col])
        for i in 1:m
            i == row && continue
            M[i, :] = simplify.(M[i, :] .- M[i, col] .* M[row, :])
        end
        push!(pivots, col)
        row += 1
    end
    all(isz, M[row:end, end]) || error("the boundary conditions are inconsistent")
    length(pivots) == n || error("$(n - length(pivots)) coefficient(s) left undetermined")
    return Dict(unknowns[pivots[i]] => M[i, end] for i in eachindex(pivots)), m - length(pivots)
end
```

4. Trial stress function: the axisymmetric terms of the Michell solution

The Michell solution (Sadd 8.3.6) is the complete list of separable solutions of the polar biharmonic equation. The terms whose **stresses** do not depend on θ are

$$\phi = a_0 + a_1 \ln r + a_2 r^2 + a_3 r^2 \ln r + a_4\, \theta$$

a_0 produces no stress and is dropped. The last term deserves a comment: $\phi = a_4\theta$ is not single-valued on a full ring, but its stresses are ($\tau_{r\theta} = a_4/r^2$, a pure axisymmetric shear — Sadd §8.3.2), and on our half ring θ is a perfectly good function. Sadd solves Example 8-6 having already discarded a_3 and a_4: a_4 because there is no shear load, a_3 because on a *full* ring its tangential displacement is multivalued (it grows by $4\pi a_3 r/E$ every time round the ring). Our domain does **not** encircle the origin, so neither argument is available to us a priori. We keep all four terms and let the boundary conditions decide.

$$\mathtt{a1}\ \log(r) + \mathtt{a4}\,\theta + r^2\,\mathtt{a2} + r^2\,\mathtt{a3}\ \log(r)$$

```
begin
    @variables a1 a2 a3 a4
    ϕ_trial = a1*log(r) + a2*r^2 + a3*r^2*log(r) + a4*θ
end
```

Biharmonic check. Every term of the Michell solution is biharmonic by construction, so this should come out zero without any condition on the coefficients:

$\nabla^4\phi =$

$$0$$

```
∇⁴ϕ = biharmonic_polar(ϕ_trial)
```

Stresses from the trial function — Sadd (8.3.8) plus the a_4 shear:

$$\left(\frac{\mathtt{a1} + 2\,r^2\,\mathtt{a2} + r^2\,\mathtt{a3} + 2\,r^2\,\mathtt{a3}\ \log(r)}{r^2},\ \frac{-\mathtt{a1} + 2\,r^2\,\mathtt{a2} + 3\,r^2\,\mathtt{a3} + 2\,r^2\,\mathtt{a3}\ \log(r)}{r^2},\ \frac{\mathtt{a4}}{r^2}\right)$$

```
σr_trial, σθ_trial, τrθ_trial = stresses_polar(ϕ_trial)
```

$$\sigma_r = \frac{a_1}{r^2} + 2a_2 + a_3\,(1 + 2\ln r), \qquad \sigma_\theta = -\frac{a_1}{r^2} + 2a_2 + a_3\,(3 + 2\ln r), \qquad \tau_{r\theta} = \frac{a_4}{r^2}$$

Plane strain strains of the trial field:

$$\frac{\mathtt{a1} + \mathtt{a1}\,\nu + 2\,r^2\,\mathtt{a2} + r^2\,\mathtt{a3} - 2\,r^2\,\mathtt{a2}\,\nu + 2\,r^2\,\mathtt{a3}\,\log(r) - 3\,r^2\,\mathtt{a3}\,\nu - 4\,\nu^2\,r^2\,\mathtt{a2} - 2\,r^2\,\mathtt{a3}\,\log(r)\,\nu - 4\,\nu^2\,r}{r^2\,E}$$

```
begin
    εr_trial  = simplify((1 + ν)/E * ((1 - ν)*σr_trial - ν*σθ_trial))
    εθ_trial  = simplify((1 + ν)/E * ((1 - ν)*σθ_trial - ν*σr_trial))
    γrθ_trial = simplify(2(1 + ν)/E * τrθ_trial)
    (εr_trial, εθ_trial, γrθ_trial)
end
```

5. Displacements of the general axisymmetric field

Before imposing any boundary condition, integrate the strain–displacement relations for the trial field. The algebra is short enough to follow by hand, with `Symbolics.jl` doing the bookkeeping.

Step 1 — radial strain. $\partial u_r/\partial r = \varepsilon_r(r)$ gives $u_r = \int \varepsilon_r\,dr + f(\theta)$, where the "constant" of integration may depend on θ:

ur_part =

$$\frac{\mathtt{a1} + \mathtt{a1}\,\nu}{-E\,r} + \frac{2\,\mathtt{a2}\,r + \mathtt{a3}\,r - 2\,\mathtt{a2}\,r\,\nu - 3\,\mathtt{a3}\,r\,\nu - 4\,\nu^2\,\mathtt{a2}\,r - 4\,\nu^2\,\mathtt{a3}\,r}{E} + \frac{-2\,\mathtt{a3}\,r + 2\,\mathtt{a3}\ \mathrm{lo}}{}$$

```
ur_part = ∫r(εr_trial)
```

Step 2 — hoop strain. $u_r + \partial u_\theta/\partial\theta = r\,\varepsilon_\theta$, so $\partial u_\theta/\partial\theta = r\varepsilon_\theta - \int \varepsilon_r\,dr - f(\theta)$. The combination $r\varepsilon_\theta - \int \varepsilon_r\,dr$ turns out to be remarkably simple — everything except the a_3 term cancels:

κr =

$$\frac{-4\,\mathtt{a3}\,r + 4\,\nu^2\,\mathtt{a3}\,r}{-E}$$

```
κr = simplify(expand(r*εθ_trial - ur_part))
```

so $\partial u_\theta/\partial\theta = \kappa r - f(\theta)$ with $\kappa = 4(1-\nu^2)a_3/E$, and $u_\theta = \kappa\,r\,\theta - F(\theta) + g(r)$ with $F' = f$.

Step 3 — shear strain. Substitute both displacements into $\gamma_{r\theta} = \frac{1}{r}\partial_\theta u_r + \partial_r u_\theta - u_\theta/r$. The $\kappa r\theta$ terms cancel ($\partial_r(\kappa r\theta) = \kappa r\theta/r$) and what is left is

$$\frac{f'(\theta) + F(\theta)}{r} + g'(r) - \frac{g(r)}{r} = \gamma_{r\theta} = \frac{2(1+\nu)\,a_4}{E\,r^2}$$

The left side separates into a function of θ and a function of r, so each must be constant.
$F'' + F = \text{const}$ gives $f = u_{x0}\cos\theta + u_{y0}\sin\theta$: a rigid translation (u_{x0}, u_{y0}) written in polar components. $g' - g/r = 2(1+\nu)a_4/(Er^2) - \text{const}/r$ gives $g = -(1+\nu)a_4/(Er) + \omega_0 r$: the displacement of the a_4 shear field plus a rigid rotation ω_0. Altogether,

$$u_r = \int \varepsilon_r\, dr + u_{x0}\cos\theta + u_{y0}\sin\theta, \qquad u_\theta = \frac{4(1-\nu^2)a_3}{E}\, r\theta - \frac{(1+\nu)a_4}{E\,r} - u_{x0}\sin\theta + u_{y0}$$

This is Sadd (8.3.9) converted to plane strain with Table 7-1, plus the a_4 term. Assemble it and let `Symbolics` confirm that it reproduces all three strains (each residual should be zero):

▸ $\left(\frac{\mathrm{a1} + \mathrm{a1}\,\nu}{-E\,r} + \frac{2\,\mathrm{a2}\,r + \mathrm{a3}\,r - 2\,\mathrm{a2}\,r\,\nu - 3\,\mathrm{a3}\,r\,\nu - 4\,\nu^2\,\mathrm{a2}\,r - 4\,\nu^2\,\mathrm{a3}\,r}{E} + \frac{-2\,\mathrm{a3}\,r + 2\,\mathrm{a3}\,\log(r)\,r + 2\,\mathrm{a3}\,r}{}\right.$

```
begin
    @variables ux0 uy0 ω0
    ur_general = ur_part + ux0*cos(θ) + uy0*sin(θ)
    uθ_general = 4(1 - ν^2)*a3*r*θ/E - (1 + ν)*a4/(E*r) - ux0*sin(θ) + uy0*cos(θ) + ω0*r
    (ur_general, uθ_general)
end
```

`strain_check_general` = ▸ $[0, 0, 0]$

```
strain_check_general = simplify.(expand.([
    expand_derivatives(Dr(ur_general)) - εr_trial,
    expand_derivatives((ur_general + Dθ(uθ_general))/r) - εθ_trial,
    expand_derivatives(Dθ(ur_general)/r + Dr(uθ_general) - uθ_general/r) - γrθ_trial,
]))
```

6. What the supports say

Impose the support and symmetry conditions on the general field. Each must hold **for every r** across the foot, so we multiply through by r (or r^2) to clear the $1/r$ terms and match coefficients of powers of r:

- right foot: $u_\theta(r, 0) = 0$ and $\tau_{r\theta}(r, 0) = 0$;
- left foot: $u_\theta(r, \pi) = 0$ and $\tau_{r\theta}(r, \pi) = 0$;
- axis of symmetry: $u_\theta(r, \pi/2) = 0$.

(`at_angle` substitutes the exact values of $\sin\theta$ and $\cos\theta$ before replacing θ itself, so no floating-point $\sin\pi = 10^{-16}$ noise enters the exact row reduction. π is kept as a symbol.)

$$\left[\frac{-\mathbf{a4}-\mathbf{a4}\cdot\nu}{E}=0,\ \omega\mathbf{0}=0,\ \mathbf{uy0}=0,\ \mathbf{a4}=0,\ \frac{-\mathbf{a4}-\mathbf{a4}\cdot\nu}{E}=0,\ \frac{4\cdot\mathbf{a3}\cdot\pi-4\cdot\nu^2\cdot\mathbf{a3}\cdot\pi}{E}+\omega\mathbf{0}=0,\ -\mathbf{uy}\right.$$

```
begin
    at_angle(expr, θ₀, sinθ₀, cosθ₀) = substitute(substitute(expr, Dict(sin(θ) =>
    sinθ₀, cos(θ) => cosθ₀)), Dict(θ => θ₀))
    π_ = Num(π)

    support_eqs = vcat(
        # right foot θ = 0, for all r:  uθ = 0,  τrθ = 0
        [e ~ 0 for e in coeffs_in(r   * at_angle(u0_general, 0, 0, 1), r)],
        [e ~ 0 for e in coeffs_in(r^2 * at_angle(τrθ_trial,  0, 0, 1), r)],
        # left foot θ = π, for all r:  uθ = 0,  τrθ = 0
        [e ~ 0 for e in coeffs_in(r   * at_angle(u0_general, π_, 0, -1), r)],
        [e ~ 0 for e in coeffs_in(r^2 * at_angle(τrθ_trial,  π_, 0, -1), r)],
        # axis of symmetry θ = π/2, for all r:  uθ = 0
        [e ~ 0 for e in coeffs_in(r   * at_angle(u0_general, π_/2, 1, 0), r)],
    )
end
```

(Dict(a4 a3 ux0 uy0 ω0), 6)

$$\Rightarrow 0,\ \Rightarrow 0,\ \Rightarrow 0,\ \Rightarrow 0,\ \Rightarrow 0$$

```
support_map, n_redundant_support = solve_linear_bcs(support_eqs, [a3, a4, ux0, uy0, ω0])
```

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equations, 6 of them redundant, and every one of the five unknowns is zero. The rigid-body motion is gone, $a_4 = 0$ because the feet carry no shear, and — the interesting one — $a_3 = 0$.

The $r^2 \ln r$ term is killed not by single-valuedness, as it is in the full ring, but by the **supports**: its tangential displacement $4(1-\nu^2)a_3 r\theta/E$ grows linearly with θ, so if the right foot is held at its level the left foot would rise or drop by $4\pi(1-\nu^2)a_3 r/E$. Physically, $a_3 \neq 0$ is the self-stress of an arch that has been forced open or closed by a prescribed angle at the feet (Sadd Exercise 8-21, the "dislocation" solution) — a stress field that exists with no pressure at all. Both feet held at the same level means no such opening, hence $a_3 = 0$. Had the supports been at different levels, or had one foot been forced to rotate, this term would be exactly what we needed.

7. What the pressure says

With $a_3 = a_4 = 0$ only the Lamé terms $a_1 \ln r + a_2 r^2$ survive. Two pressure conditions for two unknowns; the shear conditions on both surfaces are satisfied identically:

▸ $\left[\frac{\mathtt{a1} + 2 \cdot R^2 \cdot \mathtt{a2}}{R^2} = 0,\ \frac{\mathtt{a1} + 2 \cdot (R+t)^2 \cdot \mathtt{a2}}{(R+t)^2} = -p,\ 0 = 0,\ 0 = 0\right]$

```
begin
    σr_1  = substitute(σr_trial,  support_map)
    τrθ_1 = substitute(τrθ_trial, support_map)
    pressure_eqs = [
        substitute(σr_1,  Dict(r => R))     ~ 0,      # inner surface free
        substitute(σr_1,  Dict(r => R + t)) ~ -p,     # ground pressure on the outer
        surface
        substitute(τrθ_1, Dict(r => R))     ~ 0,
        substitute(τrθ_1, Dict(r => R + t)) ~ 0,
    ]
end
```

▸ (Dict(a1 a2), 2)

$$\Rightarrow \frac{R^4\,p + 2\,R^3\,p\,t + t^2\,R^2\,p}{2\,R\,t + t^2},\quad \Rightarrow \frac{-(R+t)^2\,p}{2\,(R+t)^2 - 2\,R^2}$$

```
pressure_map, n_redundant_pressure = solve_linear_bcs(pressure_eqs, [a1, a2])
```

$$a_1 = \frac{p\,R^2(R+t)^2}{t\,(2R+t)}, \qquad a_2 = -\frac{p\,(R+t)^2}{2\,t\,(2R+t)}$$

($t(2R+t) = (R+t)^2 - R^2$; these are Sadd's $A = a_1$ and $B = 2a_2$ of (8.4.2) with $r_1 = R$, $r_2 = R+t$, $p_1 = 0$, $p_2 = p$.) The stress function and the stress field:

$$\frac{r^2\,R^2\,p + 2\,r^2\,R\,p\,t + t^2\,r^2\,p - 2\,R^4\,\log(r)\,p - 4\,R^3\,\log(r)\,p\,t - 2\,t^2\,R^2\,\log(r)\,p}{-4\,R\,t - 2\,t^2}$$

```
begin
    coeff_map = merge(support_map, pressure_map)
    ϕ = simplify(substitute(ϕ_trial, coeff_map))
end
```

▸ (σr = $\frac{-R^4\,p - 2\,R^3\,p\,t + r^2\,R^2\,p - t^2\,R^2\,p + 2\,r^2\,R\,p\,t + t^2\,r^2\,p}{-2\,r^2\,R\,t - t^2\,r^2}$, σθ = $\frac{R^4\,p + 2\,R^3\,p\,t + r^2\,R^2\,p + t^2\,R^2\,p}{-2\,r^2\,R\,t - t^2\,r^2}$

```
begin
    σr, σθ, τrθ = stresses_polar(ϕ)
    σz = simplify(ν*(σr + σθ))              # plane strain: the stress that keeps εz = 0
    (σr = σr, σθ = σθ, τrθ = τrθ, σz = σz)
end
```

Written with the common factor $K = \dfrac{p\,(R+t)^2}{t\,(2R+t)}$:

$$\sigma_r = -K\left(1 - \frac{R^2}{r^2}\right), \qquad \sigma_\theta = -K\left(1 + \frac{R^2}{r^2}\right), \qquad \tau_{r\theta} = 0, \qquad \sigma_z = -2\nu K$$

Sadd (8.4.3)–(8.4.4) with the roles of the two pressures reversed from his Figure 8-9. Observations:

- **Both in-plane stresses are compressive everywhere**, and σ_θ dominates: the pressure p is small compared with the hoop stress it produces, by the factor $\sim R/t$ familiar from thin rings.
- The **largest** stress is the hoop stress at the **inner** surface, $\sigma_\theta(R) = -2K$, on the surface that carries no load at all. The outer surface, where the pressure acts, has the smaller $|\sigma_\theta(R+t)| = K(1 + R^2/(R+t)^2)$.
- σ_z is uniform through the thickness and is the only stress that knows about the material (through ν). It is compressive: the tunnel would like to lengthen under the Poisson effect of the in-plane compression, and σ_z stops it.
- The stresses split into two classic fields: the $-K$ part is uniform hydrostatic compression (the $a_2 r^2$ term, Sadd §8.4.2 in reverse), and the $\pm K R^2/r^2$ part is the *pressurised hole* field of Sadd §8.4.1 (the $a_1 \ln r$ term) — pure shear in the r–θ plane, which is why its contributions to σ_r and σ_θ are equal and opposite.

Verify every condition we imposed (each residual should be zero):

```
bc_residuals = ▸[0, 0, 0, 0, 0]
```

```
bc_residuals = simplify.(expand.([
    biharmonic_polar(ϕ),
    substitute(σr,  Dict(r => R)),
    substitute(σr,  Dict(r => R + t)) + p,
    τrθ,                                                  # for all r, θ — covers both surfaces and both feet
    substitute(ν*(σr + σθ) - σz, Dict(r => R)),           # σz is the plane strain value
]))
```

Do the supports carry the load? We never prescribed the reaction. The vertical force per unit length of tunnel that each foot delivers is the integral of the hoop stress across the foot, $N = \int_R^{R+t} \sigma_\theta(r, 0)\, dr$. Statics of the half ring says the total downward load is the vertical resultant of a radial pressure p acting on a semicircle of radius $R + t$, which is $2p(R + t)$, so each foot must carry $p(R + t)$:

▸ (N_foot = $-R\,p - p\,t$, equilibrium_residual = $(R+t)\,p - R\,p - p\,t$, M_foot = $\frac{2\,R^3\,p\,t + 3\,t^2\,R^2\,p + t^3\,l}$

```
begin
    N_foot = simplify(∫r(σθ, R, R + t))                          # hoop force per
    unit length (compression negative)
    M_foot = simplify(∫r(σθ*(r - (R + t/2)), R, R + t))          # its moment about
    mid-thickness
    (N_foot = N_foot, equilibrium_residual = simplify(N_foot + p*(R + t)), M_foot =
    M_foot)
end
```

$N = -p(R+t)$ exactly: equilibrium is built into the Airy stress function, so the reaction took care of itself. The moment of the hoop stress about the mid-thickness is not zero, because σ_θ is larger at the inner face — the resultant acts slightly *inside* mid-thickness. The thrust in the arch is therefore eccentric, by a small amount that a thin-ring analysis cannot see.

8. Strains and displacements

Plane strain Hooke's law for the final stresses, then the displacements from the general field of Section 5 with every coefficient substituted:

▸ (εr = $\frac{R^4\,p + 2\,R^3\,p\,t - r^2\,R^2\,p + t^2\,R^2\,p - 2\,r^2\,R\,p\,t - t^2\,r^2\,p + R^4\,p\,\nu + 2\,R^3\,p\,t\,\nu + r^2\,R^2\,p\,\nu + t^2\,R^2\,p\,\nu}{2\,r^2\,E\,R\,t + t^2\,r^2\,E}$

```
begin
    εr  = simplify((1 + ν)/E * ((1 - ν)*σr - ν*σθ))
    εθ  = simplify((1 + ν)/E * ((1 - ν)*σθ - ν*σr))
    γrθ = simplify(2(1 + ν)/E * τrθ)
    εz  = 0                                          # by definition of plane strain
    (εr = εr, εθ = εθ, γrθ = γrθ, εz = εz)
end
```

▸ (ur = $\frac{-R^4\,p - 2\,R^3\,p\,t - r^2\,R^2\,p - t^2\,R^2\,p - 2\,r^2\,R\,p\,t - t^2\,r^2\,p - R^4\,p\,\nu - 2\,R^3\,p\,t\,\nu + r^2\,R^2\,p\,\nu - t^2\,R^2\,p}{2\,E\,R\,r\,t + t^2\,E\,r}$

```
begin
    ur = simplify(substitute(ur_general, coeff_map))
    uθ = simplify(substitute(uθ_general, coeff_map))
    (ur = ur, uθ = uθ)
end
```

$$u_r = -\frac{(1+\nu)\,K}{E}\left[(1-2\nu)\,r + \frac{R^2}{r}\right], \qquad u_\theta = 0, \qquad u_z = 0$$

(Sadd 8.4.5 with $A = a_1$, $B = 2a_2$.) Every point moves **radially toward the centre**: the crown comes down, the haunches move in, and the feet slide horizontally inward on their smooth supports by

$|u_r(r,0)|$ — the sliding our support idealisation allows. Tunnel engineers call the resulting decrease in diameter, $2|u_r(R)|$, the *convergence* of the lining.

The two terms are the displacements of the two stress fields identified above: $(1-2\nu)r$ is the uniform contraction under hydrostatic compression, with the plane strain bulk factor $(1+\nu)(1-2\nu)/E$ (it vanishes for an incompressible material, $\nu = 1/2$, as it must); R^2/r is the volume-preserving $1/r$ displacement of the pressurised-hole field (Sadd 8.4.8).

Check that the displacements reproduce the strains:

```
strain_check = ▸[0, 0, 0]
```

```
strain_check = simplify.(expand.([
    expand_derivatives(Dr(ur)) - εr,
    expand_derivatives((ur + Dθ(uθ))/r) - εθ,
    expand_derivatives(Dθ(ur)/r + Dr(uθ) - uθ/r) - γrθ,
]))
```

Plane stress, for comparison. Had we (wrongly) treated the tunnel as a thin slice in plane stress, the stresses σ_r, σ_θ would be identical, σ_z would be zero, and the radial displacement would be Sadd (8.3.9) with $a_3 = 0$:

$$u_r^{\text{plane stress}} = \frac{1}{E}\left[2(1-\nu)\,a_2\,r - (1+\nu)\frac{a_1}{r}\right]$$

The plane strain constraint $\varepsilon_z = 0$ makes the lining stiffer in its own plane, so $|u_r|$ is smaller in plane strain. We will quantify the difference in the numerical example.

```
ur_plane_stress =
```

$$\frac{-R^4\,p - 2\,R^3\,p\,t - r^2\,R^2\,p - t^2\,R^2\,p - 2\,r^2\,R\,p\,t - t^2\,r^2\,p - R^4\,p\,\nu - 2\,R^3\,p\,t\,\nu + r^2\,R^2\,p\,\nu}{2\,E\,R\,r\,t + t^2\,E\,r}$$

```
ur_plane_stress = simplify(substitute((2(1 - ν)*a2*r - (1 + ν)*a1/r)/E, coeff_map))
```

9. Comparison with the thin-ring formula

Strength of materials treats the lining as a thin ring: the uniform external pressure p acting at radius $R + t$ produces a hoop compression $N = -p(R+t)$ per unit length (the same equilibrium we just verified), spread *uniformly* over the thickness,

$$\sigma_\theta^{\text{thin}} = \frac{N}{t} = -\frac{p\,(R+t)}{t}$$

and, neglecting σ_r (it is at most p, small compared with σ_θ), the plane strain hoop strain $\varepsilon_\theta = (1-\nu^2)\sigma_\theta/E$ at the mean radius $r_m = R + t/2$ gives a radial displacement

$$u_r^{\text{thin}} = r_m\,\varepsilon_\theta = -\frac{(1-\nu^2)\,p\,(R+t)\,(R+t/2)}{E\,t}$$

Form the differences symbolically. The *average* of the exact hoop stress through the thickness is N/t exactly — the thin-ring stress is the mean of the true one — and the exact stress at the inner and outer faces straddles it:

▸ (mean_exact_minus_thin = 0, inner_minus_thin = $\frac{R\,p + p\,t}{-2\,R - t}$, outer_minus_thin = $\frac{R\,p}{2\,R + t}$, ratio_inner

```
begin
    σθ_thin = -p*(R + t)/t
    ur_thin = -(1 - ν^2)*p*(R + t)*(R + t/2)/(E*t)
    thin_ring = (
        mean_exact_minus_thin = simplify(N_foot/t - σθ_thin),
        inner_minus_thin      = simplify(substitute(σθ, Dict(r => R)) - σθ_thin),
        outer_minus_thin      = simplify(substitute(σθ, Dict(r => R + t)) - σθ_thin),
        ratio_inner_to_outer  = simplify(substitute(σθ, Dict(r => R)) / substitute(σθ,
        Dict(r => R + t))),
    )
end
```

$$\sigma_\theta(R) - \sigma_\theta^{\text{thin}} = -\frac{p\,(R+t)}{2R+t}, \qquad \sigma_\theta(R+t) - \sigma_\theta^{\text{thin}} = +\frac{p\,R}{2R+t}$$

Both deviations are of the order of the pressure p itself, i.e. smaller than the hoop stress by a factor $\sim t/(2R)$ — the thin-ring formula is good to a few percent whenever the lining is thin. The inner-to-outer ratio depends only on t/R:

$$\frac{\sigma_\theta(R)}{\sigma_\theta(R+t)} = \frac{2(R+t)^2}{(R+t)^2 + R^2}$$

For a thin lining it tends to 1 — uniform hoop stress, the thin-ring picture. For a very thick one ($t \gg R$) it tends to 2: the stress concentration factor of a hole in an infinite medium under all-round pressure (Sadd 8.4.10 with the sign of T reversed). The lining in the example below sits much closer to the first limit.

10. Numerical example

A cast-in-place **concrete** lining: inner radius $R = 3$ m, thickness $t = 300$ mm, $E = 30$ GPa, $\nu = 0.2$. The ground pressure is taken as $p = 0.5$ MPa — roughly 25 m of soil at 20 kN/m³, applied uniformly all round. Units: N, mm, MPa; all forces are per mm of tunnel length.

```
prm = (p = 0.5, R = 3000.0, t = 300.0, E = 30000.0, ν = 0.2)
```

```
prm = (p = 0.5,          # MPa  ground pressure
       R = 3000.0,       # mm   inner radius
       t = 300.0,        # mm   lining thickness
       E = 30_000.0,     # MPa  concrete
       ν = 0.2)
```

```
(K = 2.88095, σθ_thin = -5.5, ur_thin = -0.5544)
```

```
begin
    # build_function compiles each symbolic expression to a callable f(r, p, R, t, E, ν)
    args = (r, p, R, t, E, ν)
    σr_f   = build_function(σr,   args...; expression = Val(false))
    σθ_f   = build_function(σθ,   args...; expression = Val(false))
    σz_f   = build_function(σz,   args...; expression = Val(false))
    εr_f   = build_function(εr,   args...; expression = Val(false))
    εθ_f   = build_function(εθ,   args...; expression = Val(false))
    ur_f   = build_function(ur,   args...; expression = Val(false))
    urps_f = build_function(ur_plane_stress, args...; expression = Val(false))
    ev(f, ri) = f(ri, prm.p, prm.R, prm.t, prm.E, prm.ν)
    Sr(ri) = ev(σr_f, ri);  Sθ(ri) = ev(σθ_f, ri);  Sz(ri) = ev(σz_f, ri)
    Er(ri) = ev(εr_f, ri);  Eθ(ri) = ev(εθ_f, ri)
    Ur(ri) = ev(ur_f, ri);  Urps(ri) = ev(urps_f, ri)
    σθ_thin_num = -prm.p*(prm.R + prm.t)/prm.t
    ur_thin_num = -(1 - prm.ν^2)*prm.p*(prm.R + prm.t)*(prm.R + prm.t/2)/(prm.E*prm.t)
    K_num = prm.p*(prm.R + prm.t)^2/(prm.t*(2prm.R + prm.t))
    (K = K_num, σθ_thin = σθ_thin_num, ur_thin = ur_thin_num)
end
```

Exact solution vs thin ring, side by side

$r_i = R$ is the inner (free) surface, $r_o = R + t$ the outer (loaded) one, r_m the mid-thickness.

comparison =

quantity	elasticity (plane strain)	thin ring
σθ inner surface r_i [MPa]	-5.7619	-5.5
σθ mid-thickness r_m [MPa]	-5.4941	-5.5
σθ outer surface r_o [MPa]	-5.2619	-5.5
σθ average through thickness [MPa]	-5.5	-5.5
σr inner surface r_i [MPa]	0.0	0.0
σr outer surface r_o [MPa]	-0.5	0.0
σz (uniform) [MPa]	-1.1524	-1.1
u_r inner surface r_i [mm]	-0.5531	-0.5544
u_r outer surface r_o [mm]	-0.5425	-0.5544
u_r inner surface, plane stress [mm]	-0.5762	–
convergence (decrease in inner diameter) [mm]	1.1063	1.1088
εθ inner surface [10^{-6}]	-184.381	-176.0
εr inner surface [10^{-6}]	46.0952	44.0

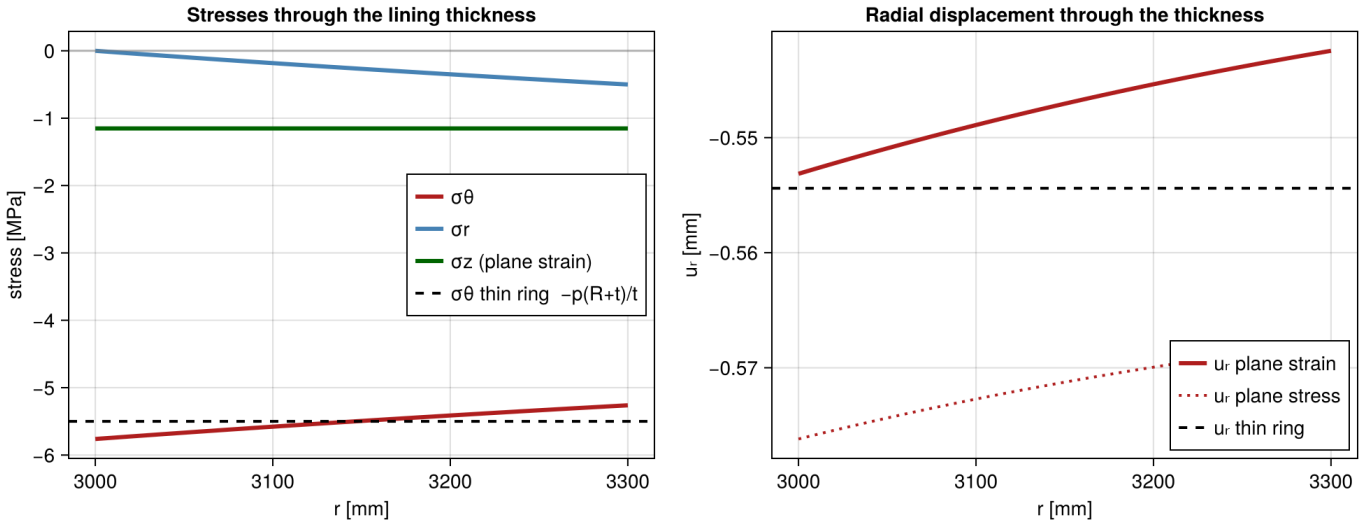
```
comparison = let
    ri, rm, ro = prm.R, prm.R + prm.t/2, prm.R + prm.t
    rows = [
        ("σθ inner surface rᵢ  [MPa]",            Sθ(ri),           σθ_thin_num),
        ("σθ mid-thickness rₘ  [MPa]",            Sθ(rm),           σθ_thin_num),
        ("σθ outer surface rₒ  [MPa]",            Sθ(ro),           σθ_thin_num),
        ("σθ average through thickness  [MPa]",   -prm.p*ro/prm.t,  σθ_thin_num),
        ("σr inner surface rᵢ  [MPa]",            Sr(ri),           0.0),
        ("σr outer surface rₒ  [MPa]",            Sr(ro),           0.0),
        ("σz (uniform)  [MPa]",                   Sz(ri),           prm.ν*σθ_thin_num),
        ("uᵣ inner surface rᵢ  [mm]",             Ur(ri),           ur_thin_num),
        ("uᵣ outer surface rₒ  [mm]",             Ur(ro),           ur_thin_num),
        ("uᵣ inner surface, plane stress  [mm]",  Urps(ri),         NaN),
        ("convergence (decrease in inner diameter)  [mm]", 2abs(Ur(ri)),
        2abs(ur_thin_num)),
        ("εθ inner surface  [10⁻⁶]",              1e6*Eθ(ri),       1e6*(1 -
        prm.ν^2)*σθ_thin_num/prm.E),
        ("εr inner surface  [10⁻⁶]",              1e6*Er(ri),       1e6*(-prm.ν*(1 +
        prm.ν))*σθ_thin_num/prm.E),
    ]
    fmt(v) = isnan(v) ? "–" : string(round(v, digits = 4) + 0.0)     # + 0.0 turns -0.0
    into 0.0
    hdr = "| quantity | elasticity (plane strain) | thin ring |\n|---|---|---|\n"
    Markdown.parse(hdr * join(["| $(q) | $(fmt(a)) | $(fmt(b)) |" for (q, a, b) in
    rows], "\n"))
end
```

With $t/R = 0.1$ the thin-ring hoop stress is the exact average, but it underestimates the inner-face stress by 4.8 % and overestimates the outer-face stress by 4.3 %. The radial stress never exceeds $p = 0.5$ MPa, about a tenth of the hoop stress, which is why the thin-ring theory can ignore it. The displacement predicted by the thin ring is within 0.2 % of the exact inner-surface value. Plane stress would overestimate the convergence by 4.2 % — the $\varepsilon_z = 0$ constraint matters even for $\nu = 0.2$. The

concrete stress of about 5.8 MPa is comfortably below a typical compressive strength, and the convergence of 1.11 mm on a 6 m tunnel is imperceptible.

Stresses and displacement through the thickness

```
fig_through =
```




```
fig_through = let
    rs = range(prm.R, prm.R + prm.t, length = 61)
    fig = Figure(size = (1000, 400))

    ax1 = Axis(fig[1, 1], title = "Stresses through the lining thickness",
               xlabel = "r [mm]", ylabel = "stress [MPa]")
    lines!(ax1, rs, Sθ.(rs), color = :firebrick, linewidth = 3, label = "σθ")
    lines!(ax1, rs, Sr.(rs), color = :steelblue, linewidth = 3, label = "σr")
    lines!(ax1, rs, Sz.(rs), color = :darkgreen, linewidth = 3, label = "σz (plane strain)")
    hlines!(ax1, [σθ_thin_num], color = :black, linestyle = :dash, linewidth = 2, label = "σθ thin ring  −p(R+t)/t")
    hlines!(ax1, [0.0], color = (:gray, 0.5))
    axislegend(ax1, position = :rc)

    ax2 = Axis(fig[1, 2], title = "Radial displacement through the thickness",
               xlabel = "r [mm]", ylabel = "uᵣ [mm]")
    lines!(ax2, rs, Ur.(rs), color = :firebrick, linewidth = 3, label = "uᵣ plane strain")
    lines!(ax2, rs, Urps.(rs), color = :firebrick, linewidth = 2, linestyle = :dot, label = "uᵣ plane stress")
    hlines!(ax2, [ur_thin_num], color = :black, linestyle = :dash, linewidth = 2, label = "uᵣ thin ring")
    axislegend(ax2, position = :rb)
    fig
end
```

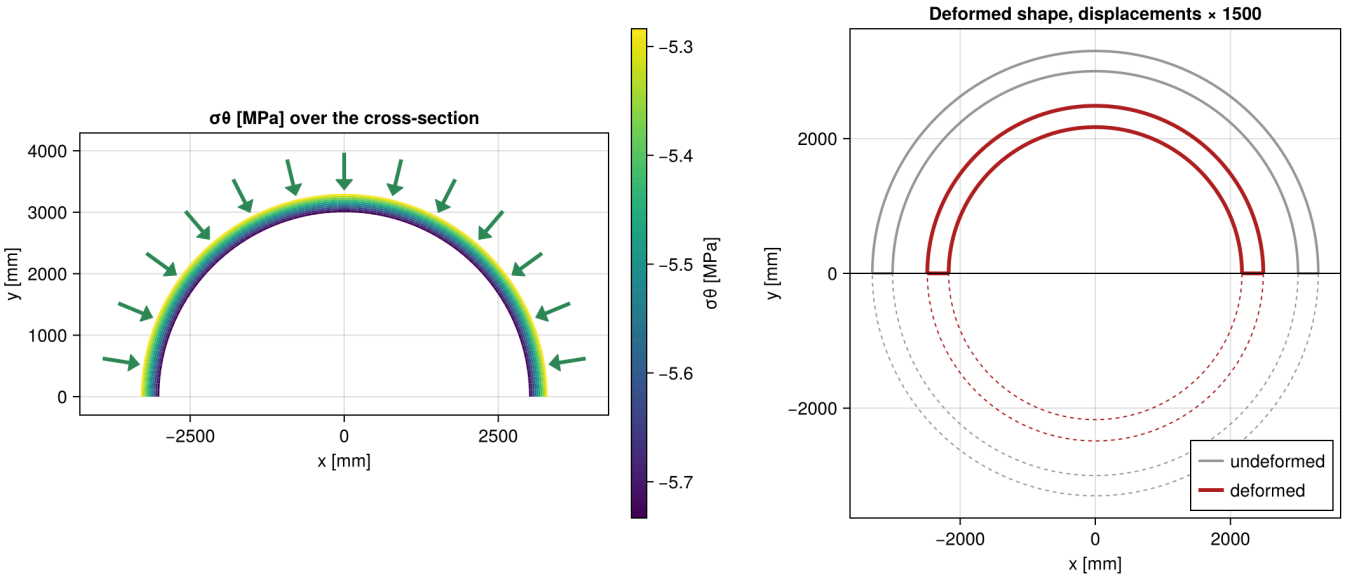
The hoop stress is largest in magnitude on the free inner face and decays as $1/r^2$ toward the outer face; the radial stress grows from zero to $-p$; σ_z is flat. The displacement is nearly uniform across the

thickness ($|u_r|$ is slightly larger inside because the inner fibre is more compressed hoop-wise), and plane strain is visibly stiffer than plane stress.

Hoop stress over the cross-section and the deformed shape

Left: σ_θ painted on the arch — axisymmetric, so it varies only with r. Right: the deformed lining with displacements exaggerated, together with the mirror image that completes the full ring of Section 2. Every point moves radially inward; the feet slide horizontally on their supports and stay at their level.

fig_section =




```julia
fig_section = let
    fig = Figure(size = (1100, 480))

    # --- hoop stress painted on the arch, as a patchwork of polar quads ---
    ax1 = Axis(fig[1, 1], title = "σθ [MPa] over the cross-section", xlabel = "x [mm]", ylabel = "y [mm]", aspect = DataAspect())
    nr, nθ = 10, 90
    redges = range(prm.R, prm.R + prm.t, length = nr + 1)
    θedges = range(0, π, length = nθ + 1)
    quads, vals = Vector{Point2f}[], Float64[]
    for i in 1:nr, j in 1:nθ
        ra, rb = redges[i], redges[i + 1]
        θa, θb = θedges[j], θedges[j + 1]
        push!(quads, Point2f[(ra*cos(θa), ra*sin(θa)), (rb*cos(θa), rb*sin(θa)), (rb*cos(θb), rb*sin(θb)), (ra*cos(θb), ra*sin(θb))])
        push!(vals, Sθ((ra + rb)/2))
    end
    poly!(ax1, quads, color = vals, colormap = :viridis, strokewidth = 0)
    Colorbar(fig[1, 2], limits = extrema(vals), colormap = :viridis, label = "σθ [MPa]")
    # pressure arrows on the outer surface
    for θa in range(0.05π, 0.95π, length = 13)
        ro = prm.R + prm.t
        arrows2d!(ax1, [1.2ro*cos(θa)], [1.2ro*sin(θa)], [-0.18ro*cos(θa)], [-0.18ro*sin(θa)], color = :seagreen)
    end
    xlims!(ax1, -1.3(prm.R + prm.t), 1.3(prm.R + prm.t)); ylims!(ax1, -0.1prm.R, 1.3(prm.R + prm.t))

    # --- deformed shape, exaggerated ---
    scale = 1500
    ax2 = Axis(fig[1, 3], title = "Deformed shape, displacements × $scale", xlabel = "x [mm]", ylabel = "y [mm]", aspect = DataAspect())
    θs = range(0, π, length = 181)
    arc(rad, s, sgn) = [Point2f((rad + s*Ur(rad))*cos(θi), sgn*(rad + s*Ur(rad))*sin(θi)) for θi in θs]

```

```
    foot(s, sgn) = [Point2f(sgn*(prm.R + s*Ur(prm.R)), 0), Point2f(sgn*(prm.R + prm.t +
    s*Ur(prm.R + prm.t)), 0)]
    for (s, col, lw, lab) in ((0, (:gray, 0.8), 2, "undeformed"), (scale, :firebrick,
    3, "deformed"))
        lines!(ax2, arc(prm.R, s, 1), color = col, linewidth = lw, label = lab)
        lines!(ax2, arc(prm.R + prm.t, s, 1), color = col, linewidth = lw)
        lines!(ax2, foot(s, 1), color = col, linewidth = lw)
        lines!(ax2, foot(s, -1), color = col, linewidth = lw)
        # mirror image: the lower half of the full ring that the smooth supports stand
        in for
        lines!(ax2, arc(prm.R, s, -1), color = col, linewidth = 1, linestyle = :dash)
        lines!(ax2, arc(prm.R + prm.t, s, -1), color = col, linewidth = 1, linestyle =
        :dash)
    end
    hlines!(ax2, [0.0], color = :black, linewidth = 1)   # the support line
    axislegend(ax2, position = :rb)
    fig
```

11. Summary

quantity	symbolic ($K = \dfrac{p(R+t)^2}{t(2R+t)}$)	numeric
ϕ	$a_1 \ln r + a_2 r^2,\quad a_1 = KR^2,\quad a_2 = -K/2$	K = 2.881 MPa
σ_r	$-K\left(1 - R^2/r^2\right)$	0 at r_i, $-p$ = -0.5 MPa at r_o
σ_θ	$-K\left(1 + R^2/r^2\right)$	-5.762 MPa at r_i, -5.262 MPa at r_o
$\tau_{r\theta}$	0	0
σ_z	$-2\nu K$	-1.152 MPa
hoop force per foot	$N = \int_R^{R+t} \sigma_\theta \, dr = -p(R+t)$	-1650.0 N/mm
u_r	$-\dfrac{(1+\nu)K}{E}\left[(1-2\nu)\,r + \dfrac{R^2}{r}\right]$	-0.5531 mm at r_i, -0.5425 mm at r_o
u_θ, u_z	0	0
thin ring σ_θ	$-p(R+t)/t$ (= exact average)	-5.5 MPa
thin ring u_r	$-(1-\nu^2)p(R+t)(R+t/2)/(Et)$	-0.5544 mm

- The **polar Airy function** turns the circular geometry into a one-dimensional problem: of the Michell terms with θ-independent stresses, $a_1 \ln r + a_2 r^2 + a_3 r^2 \ln r + a_4 \theta$, the pressure fixes a_1, a_2 and the **supports** fix $a_3 = a_4 = 0$ — the $r^2 \ln r$ "dislocation" term would lift one foot relative to the other, the θ term would shear the feet.
- Smooth supports at the feet are exactly the symmetry conditions of a **full ring** under uniform pressure, so the arch solution is Lamé's thick-walled cylinder (Sadd Example 8-6) in plane strain. What is relaxed is the friction at the feet and the non-uniformity of real ground pressure; Saint-Venant confines the first to the feet, the second needs the $\cos 2\theta$ Michell terms.
- **Plane strain** leaves the stresses σ_r, σ_θ unchanged from plane stress (traction problem, no body force), adds the uniform longitudinal stress $\sigma_z = \nu(\sigma_r + \sigma_\theta)$, and stiffens the in-plane response: the displacement factor is $(1+\nu)(1-2\nu)$ on the hydrostatic part instead of $(1-\nu)$.

- The **thin-ring** hoop stress $-p(R+t)/t$ is exactly the thickness-average of the true one; the true hoop stress is largest on the free inner face, by a factor $2(R+t)^2/[(R+t)^2+R^2]$ relative to the outer face, which ranges from 1 (thin lining) to 2 (hole in an infinite medium).